Lección 21

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Plan de trabajo

▶ **Formulación hamiltoniana**

– Ecuaciones de Hamilton-Jacobi: Examples

– Ecuaciones de Hamilton de la partícula relativista

▶ **Nuestro Syllabus:**

UNIDAD 5. Formulación Hamiltoniana

5.1. Transformaciones de Legendre

5.2. Ecuaciones de Hamilton

5.3. Ecuaciones de Hamilton de la Partícula Relativista

5.4. Corchete de Poisson

5.5. Transformaciones Canónicas

5.6. Ecuaciones de Hamilton-Jacobi

5.7. Teoría de perturbaciones

▶ **Recordatorio:**

Certamen 2: 10/06/2026

Hamilton-Jacobi equation (last lecture)

- Consider action $S = \int^{t_2} dt\, L$ as a function of final coordinates (q, p, t_2), evaluated along the classical curve

$$\frac{\partial S}{\partial q_a} = p_a, \quad \frac{\partial S}{\partial t} = -H\left(p_a,\ q_a\right), \quad \Rightarrow H\left(p_a,\ q_a\right) = H\left(\frac{\partial S}{\partial q_a},\ q_a\right) = -\frac{\partial S}{\partial t}$$

This is a **Hamilton-Jacobi equation** for unknown function $S(q_1, ..., q_N, t)$.
−A *single* equation in partial first-order derivatives.
−Expect that $S = f\left(q_1, ..., q_N; \alpha_1, ..., \alpha_N, t\right) + A$ where $\alpha_{i,}$, $A = \mathrm{const}$ which must be fixed from initial conditions

−Case of conservative systems: $E = \mathrm{const}$

$$\Rightarrow S = \underbrace{f\left(q_1, ..., q_N; \alpha_1, ..., \alpha_N\right)}_{\text{no } t-\text{dependence}} - E\, t$$

*Sometimes can find solution using **separation of variables**: $f\left(q_1, ..., q_N\right) = \sum f_a\left(q_a\right)$
To find trajectory, solve algebraic system of N equations (**Jacobi's theorem**)

$$\beta_a = \frac{\partial f}{\partial \alpha_a} = \mathrm{const}, \qquad a = 1, ..., N$$

−solution exists near any point where $\det\left(\partial^2 S/\partial q_a \partial \alpha_b\right) \neq 0$
−The β_a are constants which determine orientation in space, time, position ...

Problem 1

The lagrangian of a particle is given by

$$L = m\frac{\dot{q}_1^2 + \dot{q}_2^2}{2}\left(q_1^2 + q_2^2\right) - \frac{\alpha}{q_1^2 + q_2^2}$$

where $q_1(t)$, $q_2(t)$ are generalized coordinates and m, $\alpha \equiv \mathrm{const}$.

- Construct the generalized momenta and the hamiltonian
- Write out the canonical equations of motion
- Write out the equation of Hamilton-Jacobi and integrate it.
  Note: *The goal of this part is to test your knowledge of the Hamilton-Jacobi formalism, for this reason solution with any other method will not be taken into account.*

Problem 1 - parts 1,2

- Generalized momenta:

$$p_1 = \frac{\partial L}{\partial \dot{q}_1} = m\,\dot{q}_1\left(q_1^2 + q_2^2\right), \quad p_2 = \frac{\partial L}{\partial \dot{q}_2} = m\,\dot{q}_2\left(q_1^2 + q_2^2\right),$$

- Construct hamiltonian using Legendre transform:

$$H \equiv \sum_a p_a \dot{q}_a - L = p_1\dot{q}_1 + p_2\dot{q}_2 - L = \frac{p_1^2 + p_2^2}{2m\left(q_1^2 + q_2^2\right)} + \frac{\alpha}{q_1^2 + q_2^2} = \frac{p_1^2 + p_2^2 + 2m\alpha}{2m\left(q_1^2 + q_2^2\right)}$$

- Canonical equations:

$$\dot{p}_1 = -\frac{\partial H}{\partial q_1} = 2q_1\frac{p_1^2 + p_2^2 + 2m\alpha}{2m\left(q_1^2 + q_2^2\right)^2}, \quad \dot{p}_2 = -\frac{\partial H}{\partial q_2} = 2q_2\frac{p_1^2 + p_2^2 + 2m\alpha}{2m\left(q_1^2 + q_2^2\right)^2},$$

$$\dot{q}_1 = \frac{\partial H}{\partial p_1} = \frac{p_1}{m\left(q_1^2 + q_2^2\right)}, \quad \dot{q}_2 = \frac{\partial H}{\partial p_2} = \frac{p_2}{m\left(q_1^2 + q_2^2\right)},$$

Problem 1 - part 3

- Hamilton-Jacobi formalism:

$$\frac{\partial S}{\partial t} + H\left(p_a = \frac{\partial S}{\partial q_a},\, q_a\right) = 0$$

▶Energy is conserved since H does not depend on time
▶Apply separation of variables: effectively assume that S has a form

$$S = -Et + a\left(q_1\right) + b\left(q_2\right)$$

$$E = \frac{\left(a'\left(q_1\right)\right)^2 + \left(b'\left(q_2\right)\right)^2}{2m\left(q_1^2 + q_2^2\right)} + \frac{\alpha}{q_1^2 + q_2^2}$$
$$2m\left(q_1^2 + q_2^2\right)E = \left(a'\left(q_1\right)\right)^2 + \left(b'\left(q_2\right)\right)^2 + \alpha$$

▶ Separate terms which depend on q_1 and q_2:

$$2mq_1^2E - \left(a'\left(q_1\right)\right)^2 = \left(b'\left(q_2\right)\right)^2 - 2mEq_2^2 + \alpha$$

$-$since both parts depend on different variables q_1, q_2, they both must be constant

Problem 1

- Separate terms which depend on q_1 and q_2:

$$2mq_1^2E - \left(a'(q_1)\right)^2 = \left(b'(q_2)\right)^2 - 2mEq_2^2 + \alpha$$

► LHS and RHS depend on different variables, so equality is possible only if both are constants

$$\Rightarrow 2mq_1^2E - \left(a'(q_1)\right)^2 = \gamma, \qquad \left(b'(q_2)\right)^2 - 2mEq_2^2 + \alpha = \gamma$$

$$\Rightarrow a(q_1) = \int^{q_1} dq\,\sqrt{2mEq^2 - \gamma}, \qquad b(q_2) = \int^{q_2} dq\,\sqrt{2mEq^2 - (\alpha - \gamma)}$$

► Let's define function

$$F(q,\,c) = \int dq\sqrt{q^2 - c^2} = \frac{q}{2}\sqrt{q^2 - c^2} - \frac{a^2}{2}\ln\left(\sqrt{q^2 - c^2} + q\right)$$

Obviously

$$a(q_1) = \sqrt{2mE}\,F\left(q_1,\,\sqrt{\frac{\gamma}{2mE}}\right), \qquad b(q_2) = \sqrt{2mE}\,F\left(q_2,\,\sqrt{\frac{\alpha - \gamma}{2mE}}\right),$$

Problem 1

$$S = -Et + a(q_1) + b(q_2)$$

$$a(q_1) = \sqrt{2mE}\,F\left(q_1,\,\sqrt{\frac{\gamma}{2mE}}\right), \qquad b(q_2) = \sqrt{2mE}\,F\left(q_2,\,\sqrt{\frac{\alpha-\gamma}{2mE}}\right),$$

$$F(q,\,c) = \int dq\sqrt{q^2-c^2} = \frac{q}{2}\sqrt{q^2-c^2} - \frac{a^2}{2}\ln\left(\sqrt{q^2-c^2}+q\right)$$

► Now, after we got S, we can extract trajectories from

$$\frac{\partial S}{\partial E} = \mathrm{const} = -t_0, \qquad \frac{\partial S}{\partial \gamma} = \mathrm{const} = \beta,$$

-two algebraic equations for $q_1(t)$, $q_2(t)$

– The second equation can be solved explicitly in terms of elementary functions (though result is lengthy), it gives shape of trajectory

– The first equation cannot be solved explicitly, remains as implicit equation

Problem 1

The lagrangian of a particle is given by

$$L = m\frac{\dot{q}_1^2 + \dot{q}_2^2}{2}\left(q_1^2 + q_2^2\right) - \frac{\alpha}{q_1^2 + q_2^2}$$

►Note that after transformation

$$q_1 = \sqrt{r}\cos\varphi, \qquad q_2 = \sqrt{r}\sin\varphi,$$

the lagrangian turns into 2D motion in a Coulomb field which we studied in Lecture 5:

$$L = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) - \frac{\alpha}{r}$$

-for this case we also we got explicit trajectory but implicit time dependence.

Problem 2

2. A nonrelativistic charged particle of mass m is constrained to move in a horizontal plane, in the field of the harmonic oscillator $U = kr^2/2$ and the constant magnetic field $\boldsymbol{B}$ perpendicular to the plane. The vector potential $\boldsymbol{A}$ may be chosen as

$$\boldsymbol{A} = \frac{1}{2}\boldsymbol{B} \times \boldsymbol{r}.$$

Analyze this system in polar coordinates:

- Write out the lagrangian, the generalized momenta and construct the hamiltonian
- Write out the canonical equations of motion.
- Write out the equation of Hamilton-Jacobi and integrate it using separation of variables.
  Note: *The goal of this part is to test your knowledge of the Hamilton-Jacobi formalism, for this reason solution with any other method will not be taken into account.*

Lagrangian (see Lecture 12; in polar coordinates $\boldsymbol{v} = \dot{r}\hat{\boldsymbol{r}} + r\dot{\varphi}\hat{\boldsymbol{\varphi}} + \dot{z}\hat{\boldsymbol{z}}$):

$$L = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) - \frac{kr^2}{2} + q\underbrace{\frac{Br^2}{2}\dot{\varphi}}_{\boldsymbol{v}\cdot\boldsymbol{A}}, \qquad \boldsymbol{A} = \frac{1}{2}\boldsymbol{B}\times\boldsymbol{r} = \frac{Br}{2}\hat{\boldsymbol{\varphi}}$$

►Generalized momenta:

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r}, \quad p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = m\,r^2\dot{\varphi} + q\frac{Br^2}{2},$$

$-$no dependence on φ, cyclic variable, $p_\varphi = \mathrm{const}$

► Construct hamiltonian using Legendre transform:

$$H \equiv \sum_a p_a\dot{q}_a - L = p_r\dot{r} + p_\varphi\dot{\varphi} - L = \frac{p_r^2}{2m} + \frac{\left(p_\varphi - q\frac{Br^2}{2}\right)^2}{2mr^2} + \frac{kr^2}{2}$$

►Canonical equations:

$$\dot{p}_r = -\frac{\partial H}{\partial r} = -\frac{\left(q\frac{B}{2}\right)^2 r^4 - p_\varphi^2}{mr^3} - kr, \quad \dot{p}_\varphi = -\frac{\partial H}{\partial \varphi} = 0,$$

$$\dot{r} = \frac{\partial H}{\partial p_r} = \frac{p_1}{m}, \quad \dot{\varphi} = \frac{\partial H}{\partial p_\varphi} = \frac{\left(p_\varphi - q\frac{Br^2}{2}\right)}{m}.$$

Problem 2 - Hamilton-Jacobi formalism

$$\frac{\partial S}{\partial t} + H\left(p_a = \frac{\partial S}{\partial q_a},\, q_a\right) = 0$$

- Energy is conserved since H does not depend on time
- Apply separation of variables: effectively assume that S has a form

$$S = -Et + a(r) + b(\varphi)$$

$$E = \frac{(a'(r))^2}{2m} + \frac{\left(b'(\varphi) - q\frac{Br^2}{2}\right)^2}{2mr^2} + \frac{kr^2}{2}$$

$$2mE = (a'(r))^2 + \frac{1}{r^2}\left(b'(\varphi) - q\frac{Br^2}{2}\right)^2 + kmr^2$$

We can see that φ-dependence appears only in $b'(\varphi)$, so must conclude that

$$b'(\varphi) = p_\varphi = \mathrm{const}$$

Problem 2

$$\left(\frac{da}{dr}\right)^2 = 2mE - kmr^2 - \frac{1}{r^2}\left(p_\varphi - q\frac{Br^2}{2}\right)^2$$

$$\Rightarrow a(r) = \int^r dr\sqrt{2mE - kmr^2 - \frac{1}{r^2}\left(p_\varphi - q\frac{Br^2}{2}\right)^2}$$

$$= \int^r dr\sqrt{2m\left(E + \frac{qB}{2m}p_\varphi\right) - m\left(k + \frac{q^2B^2}{4m}\right)r^2 - \frac{p_\varphi^2}{r^2}}$$

►Same integral as for plane harmonic oscillator in polar coordinates
– but with "adjusted" energy $\tilde{E} = E + \frac{qB}{2m}p_\varphi$ and elasticity $\tilde{k} = k + \frac{q^2B^2}{4m}$
– i.e. the magnetic field just changes parameters
(Mathematica returns the result in terms of elementary functions but too lengthy)

$$\text{action}: \qquad S = -Et + a(r) + p_\varphi \varphi$$

►Now, after we got S, we can extract trajectories from

$$\frac{\partial S}{\partial E} = \text{const} = -t_0, \qquad \frac{\partial S}{\partial p_\varphi} = \text{const} = -\varphi_0.$$

"Classical" curvilinear coordinates

[Mon. Not. R. astr. Soc. (1966) 134, 253-268]

For nonrelativistic particle, the separation of variables in HJ equation is possible for 11 orthogonal coordinates, provided the potential V can be expressed in terms of arbitrary functions u, v, w of just one argument as follows:

- Cartesian x, y, z. Potential: $V = u(x) + v(y) + w(z)$
- Cylindric $\rho,\ \phi,\ z$. Potential: $V = u(\rho) + \frac{v(\phi)}{\rho^2} + w(z)$
- Spherical $r,\ \theta,\ \varphi$. Potential: $V = u(r) + \frac{v(\theta)}{r^2} + \frac{w(\varphi)}{r^2 \sin^2 \theta}$
- Parabolic cylinder ξ, η, z where ξ, η are related to Cartesian by $x = \frac{\xi - \eta}{2}$, $y = \sqrt{\xi\eta}$. Potential: $V = \frac{u(\xi)+v(\eta)}{\xi+\eta} + w(z)$. Sometimes replace $\xi \to \lambda^2$, $\eta \to \mu^2$
- Rotational parabolic coordinates ξ, η, ϕ are related to cylindric by $z = \frac{\xi - \eta}{2}$, $\rho = \sqrt{\xi\eta}$. Potential: $V = \frac{u(\xi)+v(\eta)}{\xi+\eta} + \frac{w(\phi)}{\xi\eta}$.
- Prolate and oblate spheroidal coordinates ξ, η, ϕ where ξ, η are related to cylindric by $\rho^2 = (\xi^2 \mp 1)(1 - \eta^2)$, $z = \xi\eta$.

$$\text{Potential :} \qquad V = \frac{u(\xi) + v(\eta)}{\xi^2 \mp \eta^2} + \frac{w(\phi)}{(\xi^2 \mp 1)(1 - \eta^2)}$$

- Elliptic cylinder coordinates ξ, η, z where ξ, η are related to Cartesian by $x = a\xi\eta$, $y^2 = (\xi^2 - 1)(1 - \eta^2)$. Potential: $V = \frac{u(\xi)+v(\eta)}{\xi^2 - \eta^2} + w(z)$

“Classical” curvilinear coordinates (II)

- Sphero-conal coordinates r, ξ, η, related to Cartesian by

$$x = \frac{r\mu\nu}{a\,k}, y = \frac{\sqrt{(\xi^2 - h^2)(h^2 - \eta^2)}}{k\sqrt{k^2 - h^2}}, \; z = \frac{\sqrt{(k^2 - \xi^2)(k^2 - \eta^2)}}{k\sqrt{k^2 - h^2}}, \qquad h, k = \text{const}$$

$$\text{Potential}: \quad V = w(r) + \frac{u(\xi) + v(\eta)}{r^2\,(\xi^2 - \eta^2)}$$

- Ellipsoidal coordinates λ, μ, ν related to Cartesian by

$$x^2 = \frac{(\lambda + a^2)(\mu + a^2)(\nu + a^2)}{(a^2 - b^2)(a^2 - c^2)}, \; y^2 = \frac{(\lambda + b^2)(\mu + b^2)(\nu + b^2)}{(a^2 - b^2)(b^2 - c^2)}, \; z^2 = \frac{(\lambda + c^2)(\mu + c^2)(\nu + c^2)}{(a^2 - c^2)(b^2 - c^2)}$$

$$\text{Potential}: \quad V = \frac{u(\lambda)}{(\lambda - \mu)(\lambda - \nu)} + \frac{v(\mu)}{(\mu - \nu)(\mu - \lambda)} + \frac{w(\nu)}{(\nu - \lambda)(\nu - \mu)}$$

- Paraboloidal coordinates λ, μ, ν related to Cartesian by

$$x^2 = \frac{(\lambda + a^2)(\mu + a^2)(\nu + a^2)}{(a^2 - b^2)}, \; y^2 = \frac{(\lambda + b^2)(\mu + b^2)(\nu + b^2)}{(b^2 - a^2)}, \; z^2 = \frac{\lambda + \mu + \nu - a^2 - b^2}{2}$$

$$\text{Potential}: \quad V = \frac{u(\lambda)}{(\lambda - \mu)(\lambda - \nu)} + \frac{v(\mu)}{(\mu - \nu)(\mu - \lambda)} + \frac{w(\nu)}{(\nu - \lambda)(\nu - \mu)}$$

Comment

Relation of separation of variables in Hamilton-Jacobi formalism and in Schroedinger equation in curvilinear coordinates

- Assume we have a scalar function

$$\psi = e^{iS/\hbar}, \qquad \hbar \ll 1$$

  (You may have recognized the WKB approximation from Quantum Mechanics. We won't discuss this relation in detail, just consider as purely mathematical exercise)

  - If $S = \sum s_a(q_a)$ in some curvilinear coordinates, then

$$\psi = \prod_a e^{is_a(q_a)/\hbar} = \prod_a \mathfrak{s}_a(q_a), \qquad \mathfrak{s}_a(q_a) = e^{is_a(q_a)/\hbar}$$

- Let's find the Laplacian and take the limit $\hbar \to 0$

$$-\hbar^2 \Delta\psi = \lim_{\hbar\to 0} \hbar^2 \left[\frac{(\nabla S)^2}{\hbar^2} - \frac{2i}{\hbar}\Delta S \right] \psi \approx (\nabla S)^2 \, \psi$$

$\Rightarrow$For nonrelativistic particle, the separation of variables in HJ equation of classical mechanics and in Schroedinger equation of Quantum Mechanics are closely related (can be done for the same set of potentials).

Examples

Now we'll see two examples how the separation of variables is done in curvilinear coordinates.

- We will discuss only main steps:
  - How to choose the curvilinear coordinates and write the hamiltonian
  - Demonstrate that sometimes separation of variables may become possible after algebraic manipulation
- Your mini-homework: Please try to reproduce all steps in solutions below

Example

The particle moves in the field

$$U = -\frac{\alpha}{r} + F\,z$$

(a combination of the Coulomb and homogeneous external field) . Integrate the equations of motion and find the trajectories of the particle

- *Identify the coordinates which allow to apply the separation of variables in the HJ approach. Pay attention to symmetries*
    - Can't use cylindrical coordinates, the potential is not of the form $u(\rho) + v(\phi)/\rho^2 + w(z)$, includes term

      $$-\frac{\alpha}{\sqrt{\rho^2 + z^2}}$$

      Similarly, can discard all coordinates in which the variable z doesn't mix with others (so z-dependence may appear only as $+w(z)$)
    - Can't use spherical coordinates, the potential is not of the form $u(r) + v(\theta)/r^2 + w(\phi)/r^2 \sin^2\theta$, includes the term

      $$Fz = Fr\cos\theta$$

Example

The particle moves in the field

$$U = -\frac{\alpha}{r} + F\,z$$

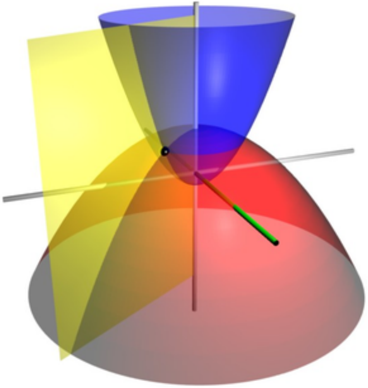
Can use parabolic coordinate system:

$$z = \frac{\xi - \eta}{2}, \qquad \rho = \sqrt{\xi\eta} = \sqrt{x^2 + y^2}, \qquad r = \sqrt{\rho^2 + z^2}$$

$$\xi = r + z, \quad \eta = r - z$$

$$z = \frac{\xi - \eta}{2}, \quad r = \frac{\xi + \eta}{2}$$

$$x = \sqrt{\xi\eta}\cos\phi, \quad y = \sqrt{\xi\eta}\sin\phi$$



Example

Lagrangian in parabolic coordinate system

$$L = \frac{m}{2}\left(\dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2\right) - U = \frac{m(\xi+\eta)}{8}\left(\frac{\dot{\xi}^2}{\xi} + \frac{\dot{\eta}^2}{\eta}\right) + \frac{m\,\xi\eta\dot{\phi}^2}{2} - U$$

$$p_\xi = \frac{m}{4}\frac{\xi+\eta}{\xi}\dot{\xi}, \quad p_\eta = \frac{m}{4}\frac{\xi+\eta}{\eta}\dot{\eta}, \quad p_\phi = m\xi\eta\dot{\phi}$$

$$H = \frac{2}{m}\frac{\xi p_\xi^2 + \eta p_\eta^2}{\xi+\eta} + \frac{p_\phi^2}{2m\xi\eta} + U$$

We'll show that separation of variables works for general potential

$$U = \frac{a(\xi) + b(\eta)}{\xi+\eta}$$

(our case corresponds to $a(\xi) = \alpha - \frac{1}{2}F\xi^2,\ b(\eta) = \alpha + \frac{1}{2}F\eta^2$)

Example

Seek solutions in the form:

$$S = f_1(\xi) + f_2(\eta) + f_3(\phi) - E\,t$$

- $p_\phi = \mathrm{const}$ since ϕ is cyclic variable $\Rightarrow f_3 = p_\phi \phi$

$$H = \frac{2}{m}\frac{\xi\left(f_1'(\xi)\right)^2 + \eta\left(f_2'(\eta)\right)^2}{\xi+\eta} + \frac{p_\phi^2}{2m\xi\eta} + \frac{a(\xi)+b(\eta)}{\xi+\eta} = E$$

We have "strange" denominators $\xi\eta$, $\xi+\eta$ which apparently do not allow to make separation of variables. What should we do ?

Example

$$H = \frac{2}{m}\frac{\xi\left(f_1'(\xi)\right)^2 + \eta\left(f_2'(\eta)\right)^2}{\xi+\eta} + \frac{p_\phi^2}{2m\xi\eta} + \frac{a(\xi)+b(\eta)}{\xi+\eta} = E$$

After multiplication by $\xi+\eta$ we end up with (below $F_{1,2}$ are known functions)

$$\left[\xi\left(f_1'(\xi)\right)^2 + F_1(\xi)\right] + \left[\eta\left(f_2'(\eta)\right)^2 + F_2(\eta)\right] = 0$$

The first term depends only on ξ, second only on η, so sum=0 only if each of them is constant ($= \pm\alpha_1$)

$$2\xi\left(f_1'(\xi)\right)^2 + \frac{p_\phi^2}{2\xi} - m\left(E\xi - a(\xi)\right) = \alpha_1$$

$$2\eta\left(f_2'(\eta)\right)^2 + \frac{p_\phi^2}{2\eta} - m\left(E\eta - b(\eta)\right) = -\alpha_1$$

Example

Formal solution:

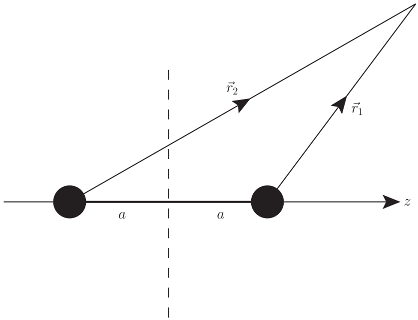
$$f_1(\xi) = \int^{\xi} \frac{d\xi}{\sqrt{2\xi}} \sqrt{\alpha_1 - \frac{p_\phi^2}{2\xi} + m\left(E\xi - a(\xi)\right)}$$

$$f_2(\eta) = \int^{\xi} \frac{d\eta}{\sqrt{2\eta}} \sqrt{-\alpha_1 - \frac{p_\phi^2}{2\eta} + m\left(E\eta - b(\eta)\right)}$$

- Free constants: $\alpha_k = (\alpha_1, E, p_\phi)$. Request $\partial S/\partial \alpha_k = \beta_k = \mathrm{const}$ to get trajectories.

$-$Our initial problem corresponds to $a(\xi) = \alpha - \frac{1}{2}F\xi^2$, $b(\eta) = \alpha + \frac{1}{2}F\eta^2$.
*Integration with *Mathematica*:
$f_{1,2} \rightarrow$elliptic functions (so end up with implicit trajectory)

Example 2



A charged particle of mass m moves in the Coulomb field of two fixed charges located at distance $2a$,

$$U = \frac{\alpha_1}{r_1} + \frac{\alpha_2}{r_2}$$

where $\alpha_{1,2} = \mathrm{const}$ and $r_{1,2} \equiv |\vec{r}_{1,2}|$ are distances from fixed points, as shown in the plot. Integrate the equations of motion for this particle.

– *What coordinates should we use for separation of variables in HJ approach ? Pay attention to symmetries of the potential*

Example 2

►We'll use elliptic coordinates $(\xi,\,\eta,\,\phi)$, which are related to the polar coordinates $(\rho, z,\,\phi)$ as

$$\rho = a\sqrt{(\xi^2-1)\,(1-\eta^2)}, \quad z = a\,\xi\eta.$$

In these coordinates the distances $r_{1,2} \equiv \sqrt{(z\mp a)^2+\rho^2}$ shown in the Figure might be rewritten as

$$r_1 = a\,(\xi-\eta)\,, \quad r_2 = a\,(\xi+\eta)$$

►Lagrangian:

$$L = T - U = \frac{m}{2}\left(\dot{\rho}^2+\rho^2\dot{\phi}^2+\dot{z}^2\right) - U =$$

$$= \frac{m\,a^2}{2}\left(\xi^2-\eta^2\right)\left(\frac{\dot{\xi}^2}{\xi^2-1}+\frac{\dot{\eta}^2}{1-\eta^2}\right) + \frac{ma^2}{2}\left(\xi^2-1\right)\left(1-\eta^2\right)\dot{\phi}^2 - \frac{\alpha_1/a}{\xi-\eta} - \frac{\alpha_2/a}{\xi+\eta}$$

Example 2

- Generalized momenta:

$$p_\xi = \frac{\partial L}{\partial \dot\xi} = m\,a^2 \left(\frac{\xi^2 - \eta^2}{\xi^2 - 1}\right)\dot\xi, \quad p_\eta = \frac{\partial L}{\partial \dot\eta} = m\,a^2 \left(\frac{\xi^2 - \eta^2}{1 - \eta^2}\right)\dot\eta,$$

$$p_\phi = \frac{\partial L}{\partial \dot\phi} = \frac{ma^2}{2}\left(\xi^2 - 1\right)\left(1 - \eta^2\right)\dot\phi$$

- Use Legendre transform to get Hamiltonian:

$$H = \dot\xi p_\xi + \dot\eta p_\xi + \dot\phi p_\phi - L =$$

$$= \frac{(\xi^2 - 1)p_\xi^2 + (1 - \eta^2)p_\eta^2}{2ma^2(\xi^2 - \eta^2)} + \frac{p_\phi^2}{2ma^2\left(\xi^2 - 1\right)\left(1 - \eta^2\right)} + \frac{\alpha_1/a}{\xi - \eta} + \frac{\alpha_2/a}{\xi + \eta}$$

Example 2

▶Hamilton-Jacobi equations:

$$\frac{\partial S}{\partial t} + H\left(p_\xi \equiv \frac{\partial S}{\partial \xi},\, p_\eta \equiv \frac{\partial S}{\partial \eta},\, p_\phi \equiv \frac{\partial S}{\partial \phi}\right) = 0,$$

$$\Downarrow$$

$$\frac{\partial S}{\partial t} + \frac{(\xi^2-1)\left(\partial_\xi S\right)^2 + (1-\eta^2)\left(\partial_\eta S\right)^2}{2ma^2(\xi^2-\eta^2)} + \frac{\left(\partial_\phi S\right)^2}{2ma^2\,(\xi^2-1)\,(1-\eta^2)} + \frac{\alpha_1/a}{\xi-\eta} + \frac{\alpha_2/a}{\xi+\eta} = 0,$$

▶ϕ is cyclic variable, $p_\phi = \mathrm{const}$▶apply separation of variables (*assume* that $F(\xi,\eta) = f_1(\xi) + f_2(\eta)$):

$$S = -Et + p_\phi \phi + f_1(\xi) + f_2(\eta). \tag{1}$$

▶HJ after substitution:

$$E = \frac{(p_\phi)^2}{2ma^2\,(\xi^2-1)\,(1-\eta^2)} + \frac{(\xi^2-1)\left(\partial_\xi f_1(\xi)\right)^2 + (1-\eta^2)\left(\partial_\eta f_2(\eta)\right)^2}{2ma^2(\xi^2-\eta^2)} + \frac{(\alpha_1+\alpha_2)\,\xi + (\alpha_1-\alpha_2)\,\eta}{a(\xi^2-\eta^2)},$$

Again, we have "strange" factors $\left(\xi^2-1\right)$, $\left(1-\eta^2\right)$, $\left(\xi^2-\eta^2\right)$ in denominator which apparently don't allow separation of variables. What should we do?

Example 2

▶HJ after substitution:

$$E = \frac{(p_\phi)^2}{2ma^2\,(\xi^2-1)\,(1-\eta^2)} + \frac{(\xi^2-1)\,(\partial_\xi f_1(\xi))^2 + (1-\eta^2)\,(\partial_\eta f_2(\eta))^2}{2ma^2(\xi^2-\eta^2)} + \frac{(\alpha_1+\alpha_2)\,\xi + (\alpha_1-\alpha_2)\,\eta}{a(\xi^2-\eta^2)},$$

▶Looks like dependence on ξ, η is not separable, but in reality it is:

– multiply both parts by $\xi^2 - \eta^2$

– take into account

$$\frac{\xi^2-\eta^2}{(\xi^2-1)\,(1-\eta^2)} = \frac{1}{\xi^2-1} + \frac{1}{1-\eta^2}$$

$$E\,\left(\xi^2 - \eta^2\right) = \frac{(\xi^2-1)\,(\partial_\xi f_1(\xi))^2}{2ma^2} + \frac{(1-\eta^2)\,(\partial_\eta f_2(\eta))^2}{2ma^2} + \frac{p_\phi^2}{2ma^2}\left(\frac{1}{\xi^2-1} + \frac{1}{1-\eta^2}\right) + \frac{(\alpha_1+\alpha_2)\,\xi + (\alpha_1-\alpha_2)\,\eta}{a},$$

-sum of the terms which depend either on ξ (in red) or on η (in blue).

–Put all ξ- and η -dependent variables in different parts of equation request that both should be a const

Example 2

$$-E\,\xi^2 + \frac{(\xi^2-1)\,(\partial_\xi f_1(\xi))^2}{2ma^2} + \frac{p_\phi^2}{2ma^2}\,\frac{1}{\xi^2-1} + \frac{(\alpha_1+\alpha_2)\,\xi}{a} = \mathrm{const} = \beta$$

$$E\,\eta^2 + \frac{(1-\eta^2)\,(\partial_\eta f_2(\eta))^2}{2ma^2} + \frac{p_\phi^2}{2ma^2}\,\frac{1}{1-\eta^2} + \frac{(\alpha_1-\alpha_2)\,\eta}{a} = -\beta = \mathrm{const}.$$

–1st order "ODEs" for functions $f_1(\xi)$, $f_2(\xi)$, integration straightforward:

–... extract $\partial_\xi f_1(\xi)$ and $\partial_\eta f_2(\eta)$, after that take integrals

$$S\,(\xi,\eta,\phi,t) = -E\,t + p_\phi\phi +$$
$$\int^\xi d\xi \left[\frac{2ma^2}{\xi^2-1}\left(E\,\xi^2 - \frac{p_\phi^2}{2ma^2}\,\frac{1}{\xi^2-1} - \frac{(\alpha_1+\alpha_2)\,\xi}{a} + \beta\right)\right]^{1/2}$$
$$+\int^\eta d\eta \left[\frac{2ma^2}{1-\eta^2}\left(-E\,\eta^2 - \frac{p_\phi^2}{2ma^2}\,\frac{1}{1-\eta^2} - \frac{(\alpha_1-\alpha_2)\,\eta}{a} - \beta\right)\right]^{1/2}$$

▶Extract trajectory as usual, taking derivatives over free parameters:

$$\frac{\partial S}{\partial \beta} = \mathrm{const}, \qquad \frac{\partial S}{\partial p_\phi} = \mathrm{const}, \qquad \frac{\partial S}{\partial E} = \mathrm{const}.$$

Summary

- Hamilton-Jacobi formalism:
  - Canonical transformation to "new" variables, where:
    - hamiltonian $\tilde{H} = 0$
    - New canonical variables $Q_a \equiv \alpha_a = \mathrm{const}$ and $P_a = \beta_a = \mathrm{const}$
    - Main equation of motion:

$$H = H(p, q) = H\left(\frac{\partial S}{\partial q_a}, q_a\right) = -\frac{\partial S}{\partial t}$$

  - Powerful tool:
    - Too sophisticated for simple problems
    - Helps to solve some nontrivial problems
    - "Standard" method: separation of variables

$$S = -Et + f_1(q_1) + f_2(q_2) + f_3(q_3)$$

(although may use any other method)